Roadmap of the project

1.Project topic:

Design a computer-vision-based pipeline to detect and track bird trajectories across diverse backgrounds, and compare it against a recent deep-learning framework (NetTrack) on the BFT dataset.

2.Goal of the project:

Motivation: Detecting and tracking animals in videos is both interesting and highly challenging. So far, we have mainly worked on image-based tasks and have not tackled video tracking problems. This project is a good opportunity to explore multi-object tracking (MOT) in the wild, and to learn how classical computer vision pipelines compare to modern deep-learning methods. We surveyed several candidate video datasets (e.g., football players, different animals) and finally chose the NetTrack paper and its BFT dataset as our main reference. NetTrack was published at CVPR 2024 and provides open-source code, pretrained weights, and datasets, so its deep-learning framework is both recent and reliable. In addition, the BFT videos are high-quality, contain diverse bird species and backgrounds, and are therefore very suitable for our project.

Technically challenging: 1.Highly dynamic and small targets 2.Complex backgrounds , change of illumination and camera motion 3.Efficiency and memory constraints 4.Multi-object tracking and ID consistency

Expected results: We will obtain MOT metrics (HOTA, MOTA, IDF1) for OpenCV and deep learning frameworks, as well as frame-level IoU between our predicted boxes and BFT ground truth boxes. We expect our framework to perform well in simpler scenes(backgrounds and illumination, but poorly in dense flocks(where birds are close too ecah other), complex scenes(backgrounds and illumination) and rapid camera motion. NetTrack's framework will outperform ours in all situations. If time permits, we will perform runtime and memory measurements to illustrate the trade-off between accuracy and efficiency between the two methods.

Research questions:

a. Can we detect and track highly dynamic small bird targets across diverse natural backgrounds using a low-memory classical computer-vision pipeline (mainly based on OpenCV)?
b. How close can such a pipeline get to the NetTrack deep-learning framework on the BFT dataset?

3.Scientific hypotheses:

We plan to test the following hypotheses:

- H1 (Feasibility of classical CV): A carefully designed classical CV pipeline (with camera motion compensation, motion-based detection, optical flow, and Kalman filtering + data association) can successfully recover a non-trivial portion of bird trajectories on the BFT validation set, achieving reasonable HOTA/MOTA/IDF1 scores despite relying only on lightweight OpenCV modules.
- H2 (Advantage of deep fine-grained tracking):
  NetTrack's fine-grained, deep-learning-based framework will significantly outperform the classical pipeline on scenes with strong dynamicity (fast motion, severe deformation, dense occlusions), because it leverages powerful object detectors and point-level

features learned from large-scale data.

- H3 (Trade-off between accuracy and efficiency):(if time allows) The classical CV pipeline will require substantially less GPU memory and may achieve faster or comparable runtime per frame on CPU, but at the cost of lower tracking accuracy compared to NetTrack. This will highlight a clear trade-off between performance and computational resources

4. Computer vision techniques used to implement:

1.Preprocessing：

- Background smoothing using bilateral filtering
- Laplacian-of-Gaussian (LoG) feature extraction
- Subtitle detection via ROI-based specular-like pixel detection with morphological operations
- Glare/specular detection using HSV thresholding and local contrast analysis
- Connected component analysis for both subtitle and glare mask refinement
- Adaptive parameter tuning based on video statistics
- Mask hole filling using Gaussian blur interpolation

2.Camera Motion Compensation：

- Phase correlation for global translation estimation
- ROI-based consensus shift estimation with weighted median and MAD outlier rejection
- Shi-Tomasi corner detection for ROI quality assessment
- ECC (Enhanced Correlation Coefficient) fallback for fine alignment
- Translation-only warping (affine model)
- Motion decision based on shift magnitude thresholding
- Specular-aware error computation with soft weighting

3.Motion-based Candidate Generation：

- Normalized intensity difference map
- LoG difference map with auto-switching logic
- KNN background subtraction with adaptive learning rates
- Multi-branch fusion (diff_n + diff_g + KNN bg)
- Morphological operations with resolution-adaptive kernels
- Connected component analysis with geometric filtering
- Nested box suppression using IoA
- Temporal persistence scoring with decay
- Background-only component suppression with persistence gates
- Edge density-based scene complexity detection

4.Candidate Refinement：

1. Reconstruct a cleaner foreground region within the ROI (post-processing)： In each bounding box's ROI, use mask_fg as the initial "foreground".
2. Delete detection boxes that are not birds (based on region geometry + specular suppression)： Perform "rule filtering" on each candidate connected component/sub-box:Delete boxes whose area/minimum side/aspect ratio is outside a reasonable range (e.g., oversized boxes, extremely thin and elongated stripes).

5.Tracking & Output：

1. Iterative Lucas–KanadeLK

2. Data association (assignment): Use IoU + center distance as the cost, combined with gating (too far/too small IoU will result in no match), and use Hungarian or greedy matching.

6．Deep Baseline (NetTrack Fine-tuning Comparison)：

To serve as a strong baseline for deep learning, comparing detection/tracking performance and speed with a pure OpenCV pipeline on the BFT value. Output: NetTrack detection/tracking results

4.Metrics to evaluate the performance:

1. Calculate the IoU between the ground truth boxes provided by BFT and our predicted boxes;
2. Calculate and compare the HOTA, MOTA, and IDF1 of our method with those using paper weights;
3. If time permits, we will compare the runtime and memory cost of both methods.

5.Sources:

Paper link: https://arxiv.org/pdf/2403.11186

Code link: https://github.com/George-Zhuang/NetTrack

Dataset link: https://drive.google.com/drive/folders/140mPnOVZY-2apH76at9yYuVGIDWOvsH_?usp=share_link

OpenCV tutorials: OpenCV official site

6.Dataset:

We used the BFT val dataset provided in the NetTrace paper as our dataset for this project. This dataset is a highly dynamic bird flock dataset, containing 6 bird species, 25 videos, and a total of 5020 frames, covering various backgrounds such as oceans, mountains, and rivers. The dataset has been manually annotated with bounding boxes and traceIDs, and these annotations are stored in JSON files in the COCO dataset format.

7.Team organization:

Member	Work
Jiren Ren	Step1、Step2、Step3
Xi Wang	Step4、Step5、Step6